Transformation Rules Patterns for RDB2RDF Views

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An *RDB2RDF view* is a triple $W = (V, S, M)$, where:
- V is the vocabulary of the *target ontology*;
- S is the *source relational schema*; and
- M is a set of *mappings* between V and S, defined by *transformation rules* (TRs).

Class Transformation Rule

A Class Transformation Rule (CTR) associated with class C has the form:

$$\psi\text{: } C(s) \leftarrow R(r), \text{URI_}R(r, s), \delta(r)$$

where:
- ψ is the TR identifier;
- C is a class in the vocabulary V;
- R is a relation in the source relational schema S;
- URI_$R(r, s)$ is the URI constructor predicate of the RDF resource corresponding to r;
- r is a tuple variable ranging over tuples of R;
- s is the URI generated derived from tuple r;
- $\delta(r)$ is an optional selection condition predicate over tuples of R.

The relation R is called the *pivot relation* of the rule, and tuples of R are called *pivot tuples*. Distinct pivot tuples must generate distinct RDF resources. Two examples of CTR:

ctr-company-1: org:Company(s) $\leftarrow$ *company*(c), URI_*company*(c, s)

ctr-company-2: foaf:Organization(s) $\leftarrow$ *company*(c), URI_*company*(c, s), $(c.type =$ "S/A")

Object Type Transformation Rule

A Object Type Transformation Rule (OTR) has the form:

$$\psi\text{: } P(s, o) \leftarrow R_D(d), B_D[d, s], R_G(g), B_G[g, o], \phi(d, g)$$

where:
- ψ is the TR identifier;
- P is an object property in V whose domain D and range G are classes in V;
- "$R_D(d), B_D[d, s]$" is the right-hand side of the CTR ψ_D that matches class D with pivot relation R_D;

- "$R_G(g), B_G[g, o]$" is the right-hand side of the CTR ψ_G that matches class G with pivot relation R_G;

- $B_D[d, s]$ and $B_G[g, o]$ are the bodies of the transformation rules formed by the URI constructor predicate and the optional conditional selection predicate related to the pivot relations R_D and R_G, respectively;

- ϕ is a relational path from R_D to R_G;

- $\phi(d, g)$ is valid if tuple d is connected to tuple g following path ϕ.

An OTR example is:

$$\textbf{otr-company-1}\text{: ex:hasEmployee}(s, o) \leftarrow \mathit{company}(c), \text{URI_}\mathit{company}(c, s), \\ \mathit{employee}(e), \text{URI_}\mathit{employee}(e, o), [\mathit{fk_company_employee}](s, o)$$

Data Type Transformation Rule

There are two patterns for Data Type Transformation Rule (DTR): *Local-DTR* and *Path-DTR*.

Local-DTR The attributes belong to the pivot relation. Its transformation rule has the following pattern:

$$\psi\text{: } P(s, v) \leftarrow R_D(d), B_D[d, s], \text{nonNull}(d.A_1), \ldots, \text{nonNull}(d.A_k), \\ \text{RDFLiteral}(d.A_1, \text{“}A_1\text{”}, R_D, v_1), \ldots, \text{RDFLiteral}(d.A_k, \text{“}A_k\text{”}, R_D, v_K), \\ T([v_1, \ldots, v_k], v)$$

where:

- ψ is the TR identifier;

- P is a datatype property in V whose domain $D \in V$;

- "$R_D(r), B_D[r, s]$" is the right-hand side for the CTR ψ_D that matches class D with pivot relation R_D;

- $A_1, \ldots, A_k$ are attributes of R_D;

- T is an optional transformation function that transforms values of attributes $A_1, \ldots, A_k$ to values of property P.

Path-DTR The attributes come from a relation associated with the pivot relation. Its transformation rule has the following pattern:

$$\psi\text{: } P(s, v) \leftarrow R_D(d), B_D[d, s], R_G(g), \text{nonNull}(g.A_1), \ldots, \text{nonNull}(g.A_k), \\ \mathit{RDFLiteral}(g.A_1, \text{“}A_1\text{”}, R_G, v_1), \ldots, \mathit{RDFLiteral}(g.A_k, \text{“}A_k\text{”}, R_G, v_k), \\ T([v_1, \ldots, v_k], v), \phi(d, g)$$

where:

- ψ is the TR identifier;

- P is a datatype property in V with domain $D \in V$;

- "$R_D(d), B_D[d, s]$" is the right-hand side for the CTR ψ_D that matches class D with pivot relation R_D;

- ϕ is a relational path from R_D to R_G;

- $A_1, \ldots, A_k$ are the attributes of R_G;

- T is an optional transformation function that transforms values of attributes $A_1, \ldots, A_k$ to values of property P.